



QUANTITATIVE METHODS 1 · CHAPTER 2

Measures of Central Tendency

Detailed solutions

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1. Mean, median and mode

Waiting times are 4, 5, 5, 6, 7, 8, 9, 26 minutes. Calculate the mode, median and mean. Which measure best describes a typical wait?

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Method. Order the data. The median is the average of positions 4 and 5 because $n=8$; the mean uses all observations. Result. Mode = 5; median = $(6+7)/2=6.5$; mean = $70/8=8.75$ minutes. Interpretation. The isolated 26-minute wait pulls the mean upward. The median is the most robust description of a typical wait, while the mean remains relevant for total capacity planning.

2. Weighted decisions

A course grade combines quiz 20%, project 30% and final examination 50%. A student scores 14, 16 and 11 out of 20.

- 1 Calculate the weighted mean.
- 2 Compare it with the unweighted mean.
- 3 Explain why the two differ.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Method. Multiply each score by its weight and verify that weights sum to one. Result. $x_w = .20(14) + .30(16) + .50(11) = 13.10$. The unweighted mean is $(14 + 16 + 11)/3 = 13.67$. Interpretation. The final examination receives the largest weight and is also the lowest score, so the weighted result is lower.

3. Grouped median and quartiles

Order values are grouped: [0,20[: 8; [20,40[: 18; [40,60[: 24; [60,100[: 10.
Estimate Q_1 , the median and Q_3 by linear interpolation.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Method. $n=60$ and cumulative counts are 8, 26, 50, 60. Use $Q=L+a(k-N_{\text{before}})/n_c$ inside the class containing target rank k . Result. Q_1 has rank 15 in [20,40[: $20+20(15-8)/18=27.78$. The median has rank 30 in [40,60[: $40+20(30-26)/24=43.33$. Q_3 has rank 45: $40+20(45-26)/24=55.83$. Interpretation. Approximately half of orders lie between EUR 27.78 and EUR 55.83, subject to the uniform-within-class approximation.

4. Geometric and harmonic means

- 1 Revenue changes by +20%, -10% and +15%. Find the equivalent average annual growth rate.
- 2 A driver covers two equal distances at 60 and 90 km/h. Find the average speed.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Method. Compound growth requires the geometric mean of growth factors; equal-distance speed requires the harmonic mean. Result. Growth factor $G=(1.20 \times .90 \times 1.15)^{1/3}=1.0754$, so average growth is about 7.54% per year. Average speed $H=2/(1/60+1/90)=72$ km/h. Interpretation. The arithmetic means, 8.33% and 75 km/h, ignore compounding and unequal travel times respectively.

5. Choosing the centre

For each case choose a principal measure and justify it: employee salaries with a few executives; most common product size; average portfolio multiplier over four years; average price paid when quantities differ; satisfaction categories from very dissatisfied to very satisfied.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Result. Salaries: median, due to right skew and outliers. Product size: mode, because the most frequent category is required. Portfolio multipliers: geometric mean, because returns compound. Average price paid: quantity-weighted arithmetic mean, $\sum p_i q_i / \sum q_i$. Satisfaction: median or mode plus the full distribution, because the scale is ordinal. Interpretation. The correct centre is determined by the question, scale and data-generating mechanism, not by habit.

6. Robust centre for delivery performance

Delivery times (minutes) for 12 orders are: 18, 19, 20, 20, 21, 21, 22, 22, 23, 24, 25, 65.

- 1 Calculate the mode, median and mean.
- 2 Recalculate the mean without the 65-minute delivery.
- 3 Recommend the principal indicator for a customer-facing dashboard and justify your choice.

Method prompt. Compare the indicators before and after removing the unusual observation; do not delete it without explanation.

Detailed solution

Step 1 — mode. The values 20, 21 and 22 each occur twice, so the distribution is multimodal.

Step 2 — median. With 12 observations, use positions 6 and 7: $(21 + 22)/2 = 21.5$ minutes.

Step 3 — mean. The total is 300, hence the mean is $300/12 = 25$ minutes. Without the unusual delivery, the total is 235 for 11 orders and the mean is $235/11 = 21.36$ minutes.

Interpretation. A single long delivery raises the mean by 3.64 minutes. The median is the best principal description of a typical customer experience, but the dashboard should also report the mean and the 65-minute incident because the latter matters for capacity and service recovery.

7. Integrated centre-selection case

A retailer reports three indicators:

Question	Data
Average basket value	120 baskets: €3,600 total
Typical annual growth	+12%, -8%, +15%, +5%
Average speed	Two equal 90-km legs at 60 and 90 km/h

- 1 Choose and calculate the appropriate mean for each row.
- 2 Explain why using the arithmetic mean in all three cases is incorrect.
- 3 Write a short managerial conclusion.

Method prompt. Match additive totals, compounded factors and equal-distance rates with arithmetic, geometric and harmonic means.

Detailed solution

Basket value. The arithmetic mean preserves the total: $3,600/120 = €30$.

Growth. Multiply the growth factors: $1.12 \times 0.92 \times 1.15 \times 1.05 = 1.244376$. The equivalent annual factor is $1.244376^{1/4} = 1.0561$, hence 5.61% per year.

Speed. Equal distances require the harmonic mean: $2/(1/60 + 1/90) = 72$ km/h.

Why the methods differ. Basket values combine additively, growth compounds multiplicatively, and equal-distance speeds imply unequal travel times. The arithmetic alternatives (6% growth and 75 km/h) do not preserve the relevant totals.

Managerial conclusion. The average basket is €30, the equivalent annual growth rate is about 5.6%, and the route's average speed is 72 km/h. Each result is valid because the averaging rule matches the business mechanism.